Gaussian Process

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1. Introduction and Motivation
2. Multivariate Gaussian Distribution
3. Multivariate Gaussian Process
4. Prior in GP
5. Marginal Likelihood in GP
6. Posterior in GP
7. Process
8. GP with Bayes

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Why Gaussian Processes?

The GP is a model of Bayesian modelling of functions. It generalizes the "prior on parameters" to the "prior on the entire function", thus perfectly embodying the Bayesian philosophy (GP = Bayesian + Kernel)

- Goal: model an unknown function $\mathrm{f(x)}$ from data $\mathcal{D} = \{(\mathrm{x_i}, \mathrm{y_i})\}_{\mathrm{i=1}}^{\mathrm{n}}$
- Challenges with parametric models: refer to Table ??.
- GP offers:
  - Distribution over functions
    Example: Instead of predicting a single best-fit line, GP gives a range of plausible functions consistent with the data.
  - Uncertainty quantification
    Example: In regions with little or no data, GP naturally increases prediction variance — helpful in scientific decision-making or active learning.

GP is a Bayesian Nonparametric Model

- Bayesian: places a prior over the unknown function:

$$\mathrm{f}(\mathrm{x}) \sim \mathcal{GP}(\mathrm{m}(\mathrm{x}), \mathrm{k}(\mathrm{x}, \mathrm{x}'))$$

- Nonparametric:
  - No fixed-dimensional parameter vector (unlike θ in parametric models)
  - Model complexity grows with data
  - Prior and posterior are over functions
- Example comparison:
  - Bayesian linear regression: $\mathrm{f}(\mathrm{x}) = \mathrm{w}^\top \mathrm{x}$, $\mathrm{w} \sim \mathcal{N}(0, \Sigma)$
  - GP: $\mathrm{f}(\mathrm{x}) \sim \mathcal{GP}(0, \mathrm{k}(\mathrm{x}, \mathrm{x}'))$, where k implicitly defines infinite-dimensional feature space
- Key intuition: GP generalizes Bayesian linear models with infinite basis functions

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Multivariate Gaussian Distribution

A multivariate Gaussian (normal) distribution is a generalisation of the 1D Gaussian to n dimensions. It describes a vector of random variables whose joint distribution is Gaussian.

- A random vector $\mathrm{x} = [\mathrm{x}_1, \ldots, \mathrm{x}_\mathrm{n}]^\top \in \mathbb{R}^\mathrm{n}$ is multivariate Gaussian if:

$$\mathrm{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

  where:
  - $\boldsymbol{\mu} \in \mathbb{R}^\mathrm{n}$ is the mean vector
  - $\Sigma \in \mathbb{R}^{\mathrm{n}\times\mathrm{n}}$ is the symmetric positive semi-definite covariance matrix
- Probability density function:

$$\mathrm{p}(\mathrm{x}) = \frac{1}{(2\pi)^{\mathrm{n}/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathrm{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathrm{x}-\boldsymbol{\mu})\right)$$

- Key properties:
  - Linearity: Any affine transformation of x is also Gaussian
    Example: If $\mathrm{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ and A is a matrix, then $\mathrm{z} = \mathrm{Ax} + \mathrm{b}$ is also Gaussian.
    Let:

$$\mathrm{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

$$\mathrm{z} = \mathrm{Ax} + \mathrm{b}, \quad \mathrm{A} \in \mathbb{R}^{\mathrm{m}}$$

    times n, $\mathrm{b} \in \mathbb{R}^{\mathrm{m}}$Then:

$$\mathrm{z} \sim \mathcal{N}(\mathrm{A}\boldsymbol{\mu} + \mathrm{b},\ \mathrm{A}\Sigma\mathrm{A}^{\mathrm{T}})$$

- Key properties:
  - Marginals: Any subset of variables is jointly Gaussian
    Example: If $\mathrm{x} = [\mathrm{x}_1, \mathrm{x}_2, \mathrm{x}_3]^\top \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, then $[\mathrm{x}_1, \mathrm{x}_3]^\top$ is also Gaussian with corresponding subvector mean and submatrix covariance.
    Suppose:

$$\mathrm{x} = \begin{bmatrix} \mathrm{x}_1 \\ \mathrm{x}_2 \\ \mathrm{x}_3 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} \end{bmatrix} \right)$$

    Then the marginal distribution of $[\mathrm{x}_1, \mathrm{x}_3]^{\mathrm{T}}$ is:

$$\begin{bmatrix} \mathrm{x}_1 \\ \mathrm{x}_3 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_3 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{13} \\ \Sigma_{31} & \Sigma_{33} \end{bmatrix} \right)$$

- Key properties:
  - Conditionals: Conditional distributions are also Gaussian Example: If $\mathrm{x} = [\mathrm{x}_1, \mathrm{x}_2]^\top$, then $\mathrm{p}(\mathrm{x}_1|\mathrm{x}_2)$ is Gaussian with analytically computable mean and covariance.

$$\mathrm{x} = \begin{bmatrix} \mathrm{x}_1 \\ \mathrm{x}_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

Then the conditional distribution is:

$$\mathrm{x}_1|\mathrm{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_{1|2}, \Sigma_{1|2})$$

where:

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathrm{x}_2 - \boldsymbol{\mu}_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

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From Multivariate Gaussian to Gaussian Process

A GP generalises the multivariate Gaussian distribution to an infinite-dimensional space. It defines a prior over functions, and any finite subset of function values follows a multivariate Gaussian distribution.

- Definition:
  A function $\mathrm{f(x)}$ is said to follow a Gaussian Process with mean function $\mathrm{m(x)}$ and covariance function $\mathrm{k(x, x')}$ if:

$$\mathrm{f(x)} \sim \mathcal{GP}(\mathrm{m(x)}, \mathrm{k(x, x')})$$

  This means that for any finite set of inputs $\{\mathrm{x_1}, \ldots, \mathrm{x_n}\}$:

$$\mathrm{f} = \begin{bmatrix} \mathrm{f(x_1)} \\ \vdots \\ \mathrm{f(x_n)} \end{bmatrix} \sim \mathcal{N}(\boldsymbol{\mu}, \mathrm{K})$$

  where:

  - $\boldsymbol{\mu}_\mathrm{i} = \mathrm{m(x_i)}$ is the mean vector
  - $\mathrm{K_{ij}} = \mathrm{k(x_i, x_j)}$ is the covariance (kernel) matrix

- Meaning:
  - Infinite interpretation (from finite to infinite):
    A GP models an entire function $f(x)$, rather than a fixed-dimensional vector.
    GP can be thought of as assigning a random variable to every input $x \in \mathcal{X}$, i.e.,
    $$f = \{f(x) \mid x \in \mathcal{X}\}$$
    This is essentially an infinite-dimensional random vector.
    - Suppose we want to model a function $f(x)$, where $x \in \mathbb{R}$ and $f(x) \in \mathbb{R}$.
    - For any finite set of inputs $x_1, \ldots, x_n$, we get a finite-dimensional output vector:
      $$f = [f(x_1), \ldots, f(x_n)]^T \in \mathbb{R}^n$$
    - But the full function f encodes infinitely many outputs:
      $$f = \{f(x) \mid x \in \mathbb{R}\}$$
    - So we can think of f as an infinite-dimensional object — a random function.

- Meaning:
  - Bayesian interpretation:
    In standard Bayesian models, we place a prior on parameters: $p(\theta)$.
    In GP, the parameter is the function f itself — so we place a prior over functions:
    $$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$
    This means that the function f is a sample from a distribution over functions.
  - Joint functions interpretation:
    A GP is a distribution over functions:
    Each f is a random function where any finite collection of outputs is multivariate Gaussian.
    - ★ A GP is defined so that for any finite collection $x_1, \ldots, x_n$, the vector:
      $$[f(x_1), \ldots, f(x_n)] \sim \mathcal{N}(\boldsymbol{\mu}, K)$$
    - ★ Therefore, GP defines not just finitely many variables, but a consistent joint distribution for all $x \in \mathcal{X}$.
    - ★ GP is an infinite-dimensional generalization of multivariate Gaussian distributions.

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GP Prior: Distribution Over Functions

A Gaussian Process defines a prior over functions, meaning before observing any data, we express our belief about what functions are likely.

- A common choice is zero mean:

  $$m(x) = 0$$

  This implies the prior expectation of function values is zero at any input, but the kernel governs variation.

- Given inputs $X = [x_1, \ldots, x_n]^T$, the prior over the vector of function values is:

  $$f(X) = [f(x_1), \ldots, f(x_n)]^T \sim \mathcal{N}(0, K(X, X))$$

  where $K(X, X)$ is the $n \times n$ kernel matrix.

- The kernel (covariance) matrix is defined by:

  $$K_{ij} = k(x_i, x_j)$$

  The choice of kernel $k(\cdot, \cdot)$ determines the smoothness, periodicity, and other structural properties of sample functions.

- Interpretation: Sampling a function from the GP prior means drawing a random function where each $f(x_i)$ is jointly Gaussian with the others.

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GP Regression Setup

In GP Regression, we observe noisy outputs $\{(x_i, y_i)\}_{i=1}^n$ and aim to infer the underlying function $f(x)$.

- Observation model:
  Each observation y_i is modeled as:
  $$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$$
  where ε_i is independent Gaussian noise with variance σ_n^2.
- Induced prior over noisy observations:
  Because $y_i = f(x_i) + \varepsilon_i$, the vector of noisy observations $y = [y_1, \ldots, y_n]^T$ has the distribution:
  $$y = f + \boldsymbol{\varepsilon} \sim \mathcal{N}(0, K + \sigma_n^2 I), \quad f \sim \mathcal{N}(0, K), \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma_n^2 I), f \perp \boldsymbol{\varepsilon}$$
  where:
  - $K \in \mathbb{R}^{n \times n}$ is the kernel matrix with $K_{ij} = k(x_i, x_j)$
  - $\sigma_n^2 I$ accounts for independent noise in each observation
- This defines the GP likelihood:
  The joint probability of observed data y under the GP prior.
  $$p(y \mid X) \sim \mathcal{N}(y;\ 0,\ K + \sigma_n^2 I)$$

Marginal Likelihood in GP Regression

The marginal likelihood (also called model evidence) measures how likely the observed data is under the GP prior. It balances model fit and complexity, and is used to learn hyperparameters.

- Data likelihood under the GP prior:
  Given:
  - Training inputs: $X = [x_1, \ldots, x_n]^T$
  - Observations: $y = [y_1, \ldots, y_n]^T$
  - $y \sim \mathcal{N}(0, K + \sigma_n^2 I)$

  The marginal likelihood is:
  $$p(y|X) \sim \mathcal{N}(y; 0, K + \sigma_n^2 I)$$
- Log marginal likelihood (LML):
  The log of the marginal likelihood (more numerically stable for optimization) is:
  $$\log p(y) = -\frac{1}{2} y^T (K + \sigma_n^2 I)^{-1} y - \frac{1}{2} \log |K + \sigma_n^2 I| - \frac{n}{2} \log 2\pi$$
- Interpretation of the three terms:
  - $-\frac{1}{2} y^T (K + \sigma_n^2 I)^{-1} y$: data fit term (how well model explains data)
  - $-\frac{1}{2} \log |K + \sigma_n^2 I|$: complexity penalty (model capacity)
  - $-\frac{n}{2} \log 2\pi$: normalization constant
- Usage: Maximize LML with respect to hyperparameters (e.g., kernel

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Posterior Distribution in GP Regression

After observing training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, we want to make predictions at new test inputs $X_* = [x_*^{(1)}, \ldots, x_*^{(m)}]^T$.

- Joint prior over training and test outputs:
  Let:
  - $y = [y_1, \ldots, y_n]^T$ (noisy observations)
  - $f_* = [f(x_*^{(1)}), \ldots, f(x_*^{(m)})]^T$ (latent function at test points)

  Then, under the GP prior:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K + \sigma_n^2 I & K_*^T \\ K_* & K_{**} \end{bmatrix}\right)$$

  where:
  - $K \in \mathbb{R}^{n \times n}$: kernel matrix over training inputs
  - $K_* \in \mathbb{R}^{m \times n}$: kernel between test and training inputs
  - $K_{**} \in \mathbb{R}^{m \times m}$: kernel over test inputs

- Posterior distribution (conditioning on y):

$$\mathrm{f}_*|\mathrm{y} \sim \mathcal{N}(\boldsymbol{\mu}_*, \Sigma_*)$$

where:

$$\boldsymbol{\mu}_* = \mathrm{K}_*(\mathrm{K} + \sigma_\mathrm{n}^2\mathrm{I})^{-1}\mathrm{y}$$
$$\Sigma_* = \mathrm{K}_{**} - \mathrm{K}_*(\mathrm{K} + \sigma_\mathrm{n}^2\mathrm{I})^{-1}\mathrm{K}_*^\mathrm{T}$$

- Interpretation: The posterior gives both:
  - Predictive mean (best guess at $\mathrm{f}(\mathrm{x}_*)$)
  - Predictive variance (uncertainty at each x_*)

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Steps for Using Gaussian Process on Real Data

Step 1: Prepare the data

- Inputs: $X = [x_1, \ldots, x_n]^T \in \mathbb{R}^{n \times d}$
- Observations: $y = [y_1, \ldots, y_n]^T \in \mathbb{R}^n$

Step 2: Choose a kernel function $k(x, x')$

- Examples:
  - RBF: $k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right)$
  - Linear: $k(x, x') = x^\top x'$
  - Periodic, Matern, etc.

Step 3: Build the covariance matrix

- Compute kernel matrix: $K_{ij} = k(x_i, x_j)$
- Add noise: $K_{\text{train}} = K + \sigma_n^2 I$

Steps for Using Gaussian Process on Real Data

Step 4: Optimize hyperparameters

- Maximize the log marginal likelihood:

$$\log p(y|X) = -\frac{1}{2} y^\top K^{-1} y - \frac{1}{2} \log |K| - \frac{n}{2} \log 2\pi$$

Step 5: Predict at new inputs x_*

- Compute:

$$K_* = k(X, x_*), \quad K_{**} = k(x_*, x_*)$$

- Predictive distribution:

$$f_* \mid y \sim \mathcal{N}(\mu_*, \Sigma_*)$$

$$\mu_* = K_*^\top (K + \sigma_n^2 I)^{-1} y$$

$$\Sigma_* = K_{**} - K_*^\top (K + \sigma_n^2 I)^{-1} K_*$$

For each new input x_*, you can not only output the predictive mean μ_*, but also obtain the corresponding predictive uncertainty Σ_*

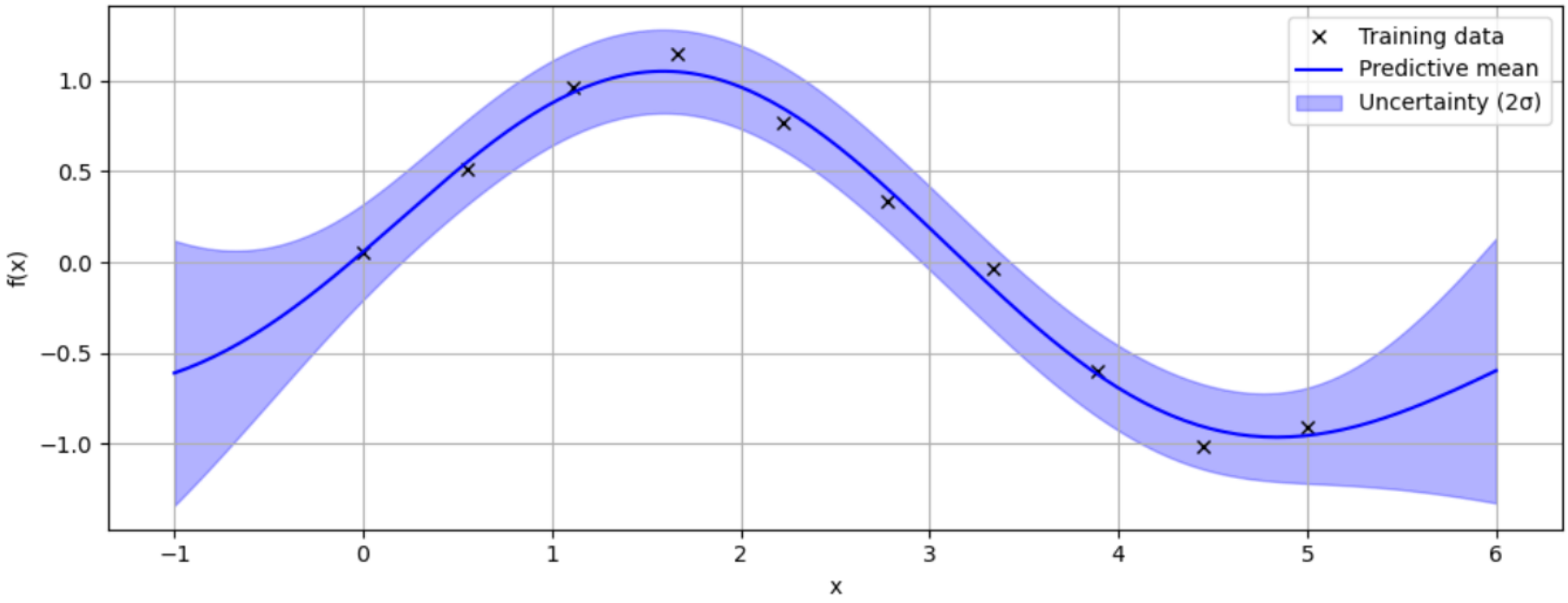

Figure: Gaussian Process Regression: Prediction with Uncertainty

- The black crosses represent the training data points (x_i, y_i).
- The blue solid line shows the predictive mean $\mu_*(x)$ of the Gaussian process.
- The light blue shaded region indicates the predictive uncertainty, corresponding to ± 2 standard deviations $\Sigma_*^{\frac{1}{2}}$ (approximately 95% confidence interval).

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GP and Bayes

These three concepts are closely related, but differ in their roles:

- Gaussian Process (GP): A nonparametric Bayesian prior over functions.
  - Defines a distribution over functions: $\mathrm{f}(\mathrm{x}) \sim \mathcal{GP}(\mathrm{m}(\mathrm{x}), \mathrm{k}(\mathrm{x}, \mathrm{x}'))$
  - Can be used for regression, classification, or as a surrogate model
- Bayesian Regression: Predict y from x with uncertainty, Output: Posterior predictive distribution $\mathrm{p}(\mathrm{y}_* \mid \mathrm{x}_*, \mathcal{D})$
  - Bayesian linear regression: prior over weights θ, $\mathrm{y} = \mathrm{x}^\top \theta + \varepsilon$
  - Gaussian Process Regression: prior over functions f, $\mathrm{y} = \mathrm{f}(\mathrm{x}) + \varepsilon$, where $\mathrm{f} \sim \mathcal{GP}$
  - GP regression is a form of Bayesian regression

Bayesian Regression vs Bayesian Optimization

- Bayesian Optimization (BO): A strategy for global optimization of black-box functions.
  - Goal: Find the input x^* that optimizes an expensive black-box function f(x)
  - Uses a surrogate model (e.g., Gaussian Process) to approximate unknown objective function f(x)
  - Uses an acquisition function to decide where to evaluate next
  - Iteratively updates the surrogate using new observations
- Summary:
  - Bayesian regression predicts outputs with uncertainty
  - Bayesian optimization finds optimal inputs with minimal evaluations

Common Acquisition Function in BO

Acquisition Function:

- A utility function that guides the selection of the next query point x.
- It balances:
  - Exploitation: Choose x where $\mu(\mathrm{x})$ is high
  - Exploration: Choose x where $\sigma(\mathrm{x})$ is high
- Example 1: Upper Confidence Bound (UCB)

$$\mathrm{UCB}(\mathrm{x}) = \mu(\mathrm{x}) + \kappa \cdot \sigma(\mathrm{x})$$

  - $\mu(\mathrm{x})$ is the surrogate's predicted mean; $\sigma(\mathrm{x})$ is its uncertainty.
  - κ is a tunable parameter controlling the exploration-exploitation trade-off.
  - Encourages selection of points with high predicted value or high uncertainty.

- Example 2: Expected Improvement (EI)
  - Formula:
$$\mathrm{EI}(x) = \mathbb{E}[\max(f(x) - f^+, 0)]$$
  - f^+ is the best observed value so far.
  - Measures the expected amount by which $f(x)$ exceeds f^+.
  - Closed-form under Gaussian assumption:
$$z = \frac{\mu(x) - f^+}{\sigma(x)}, \quad \mathrm{EI}(x) = (\mu(x) - f^+)\Phi(z) + \sigma(x)\phi(z)$$
- Example 3: Probability of Improvement (PI)
  - Formula:
$$\mathrm{PI}(x) = P(f(x) > f^+) = \Phi\left(\frac{\mu(x) - f^+}{\sigma(x)}\right)$$
  - Measures the probability that $f(x)$ improves over the current best.
  - Simpler but may be overly greedy.

Summary:
- UCB: simple, exploration-exploitation controlled via κ
- EI: balances improvement size and probability
- PI: focuses on improvement likelihood

Example: EI vs PI

Setting:

- Current best observed value: $\mathrm{f}^+ = 0.8$
- At a new input x, the surrogate predicts:

$$\mathrm{f(x)} \sim \mathcal{N}(\mu = 0.7, \sigma = 0.2)$$

1. Expected Improvement (EI):
   - EI measures the expected size of improvement:

$$\mathrm{EI(x)} = (\mu - \mathrm{f}^+)\Phi(\mathrm{z}) + \sigma\phi(\mathrm{z})$$

   - With $\mathrm{z} = -0.5$, $\Phi(-0.5) \approx 0.3085$, $\phi(-0.5) \approx 0.3521$
   - Computation:

$$\mathrm{EI(x)} = (-0.1)(0.3085) + 0.2(0.3521) = -0.0309 + 0.0704 = 0.0395$$

   - Interpretation: Though x has a 69% chance to improve, the average improvement is small.

2. Probability of Improvement (PI):
   - PI measures: $P(f(x) > f^+) = P(f(x) > 0.8)$
   - Standardized: $z = \frac{\mu - f^+}{\sigma} = \frac{0.7-0.8}{0.2} = -0.5$
   - PI:
   $$PI(x) = \Phi(0.5) \approx 0.6915$$
   - Interpretation: There is a 69% chance that f(x) improves over the current best.

Conclusion:

- PI measures likelihood of improvement
- EI considers both probability and magnitude of improvement

Bayesian Optimization using UCB

Goal: Find the global maximizer of an unknown objective function:

$$x^* = \arg\max_{x \in \mathcal{X}} f(x)$$

- f(x) is expensive, non-convex, or black-box - $\mathcal{X}$ is the candidate input space (e.g., bounded domain or discrete grid)
Bayesian Optimization Procedure:

1. Initialize: Sample initial data $\mathcal{D}_0 = \{(x_i, y_i)\}_{i=1}^n$ from f(x)
2. Fit surrogate model: Use a Gaussian Process to model f(x) with posterior:

$$f(x) \sim \mathcal{N}(\mu(x), \sigma^2(x))$$

3. Define acquisition function (UCB):

$$\mathrm{UCB}(x) = \mu(x) + \kappa \cdot \sigma(x)$$

4. Select next point:

$$x_{\text{next}} = \arg\max_{x \in \mathcal{X}} \mathrm{UCB}(x)$$

5. Evaluate objective: Observe $y_{\text{next}} = f(x_{\text{next}})$
6. Update: Add $(x_{\text{next}}, y_{\text{next}})$ to $\mathcal{D}$ and refit GP
7. Repeat steps 2–6 until convergence or budget exhausted

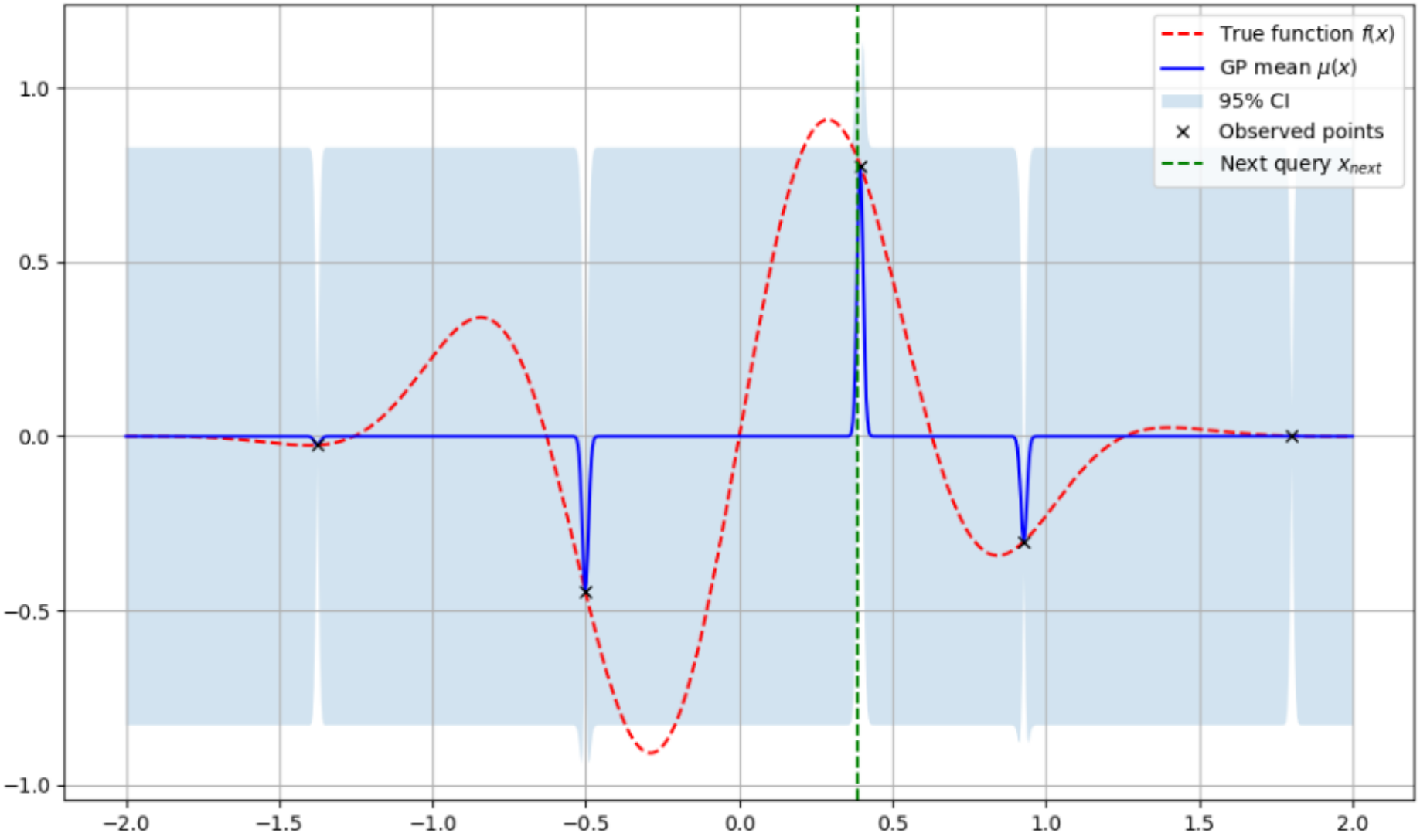

Figure: Bayesian Optimization using UCB with $\kappa = 2$.

Applications and Summary of GPs

GP can be used in any situation where traditional regression methods are applicable, often providing greater flexibility and uncertainty estimation.